How-To Library

Informatica

Converting Relational Input into Hierarchical Output using Kafka Connector

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Abstract

When you want to read data from a relational source and write data to Apache Kafka, you can use the Hierarchical Builder transformation to convert relational input into hierarchical output. This article describes how to use the Hierarchical Builder transformation with Kafka Connector to convert relational input into hierarchical output.

Supported Versions

- Informatica Cloud® Data Integration Kafka Connector

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Overview

When you want to write data from relational sources to Kafka, you can use the Hierarchical Builder transformation to convert relational input into hierarchical output.

The transformation processes a relational input from the upstream transformation and provides JSON output to the downstream transformation. Use a schema file to create a hierarchical schema that defines the expected hierarchy of the output data. Create and configure the Hierarchical Parser transformation in Data Integration and add it to a mapping. Then, associate the schema definition file with the Hierarchical Builder transformation. When you run the mapping, the Hierarchical Builder transformation converts the relational input into the hierarchical output based on the schema file.

To configure a Hierarchical Builder transformation in a Kafka mapping, perform the following steps:

1. Configure a JSON schema definition file on the Secure Agent machine.
2. Add a Source transformation.
3. Add a Hierarchical Builder transformation to the mapping.
   - Associate the JSON schema definition file with the Hierarchical Builder transformation.
   - Configure the field mapping and link the relational fields to schema elements that provide the hierarchical output.
4. Add a Target transformation.
5. Run the mapping or the mapping task.

Before You Begin

Verify that you have the permission to use the following licenses and packages before you create a mapping:

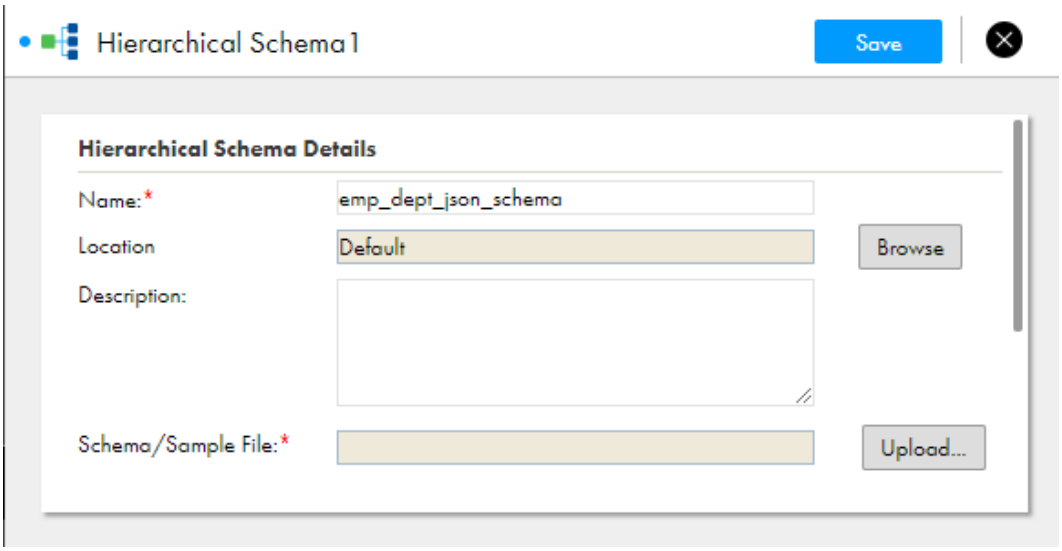
- DT XML Processing (License)
- Data Transformation
- UTD for Hierarchy

Create a Hierarchical Schema

You must associate a JSON schema definition file with the Hierarchical Builder transformation to define the schema of the Kafka target object. The Hierarchy Builder transformation converts hierarchical input based on the JSON schema definition file.

1. In Data Integration, click **New** > **Components** > **Hierarchical Schema**.

   The **Hierarchical Schema 1** page appears.



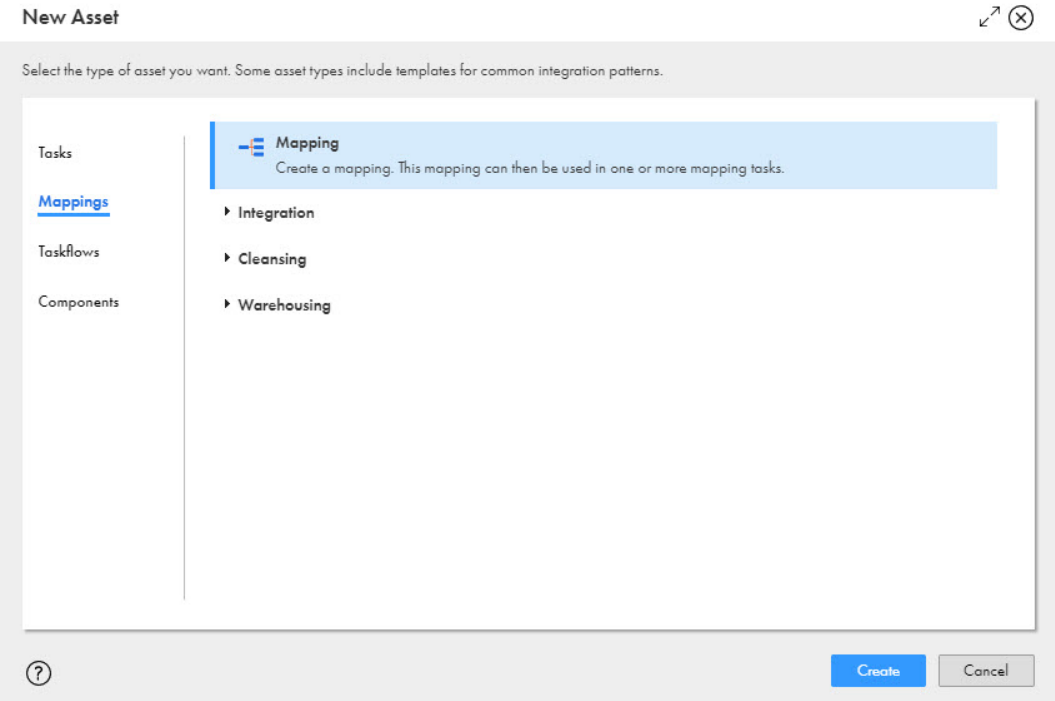
2. Enter a name for the hierarchical schema and optionally enter a description.
3. To select a schema, click **Upload**, browse for a schema file, and click **OK**.
4. To save the hierarchical schema, click **Save**.

Add a Source Transformation

Add one or more relational source objects and configure the source properties.

1. To create a mapping, click **New** > **Mapping**.

   The following image shows the **New Asset** dialog box with Mapping selected:



2. Click **Create**.

   The **Mapping Designer** appears with a new mapping displayed in the mapping canvas.
3. On the mapping canvas, click the Source transformation.
4. Specify the name and description of the source object.
5. Click **Source** and select an active Oracle connection.
6. Select `Single Object` as Source Type.
7. Configure advance properties for the source objects as required.
8. Select a Flat file as the object from where you want to read the data.
9. Repeat the steps to add more relational sources to the mapping, if required.

Note: When you read data from a Kafka topic in binary format and write the data to a target in string format, you must edit the metadata of the **data** field in the Kafka source and change the platform data type from binary to string.

Add a Hierarchical Builder Transformation to the Mapping

You must add a Hierarchical Builder transformation to the mapping to convert the relational input into hierarchical output. The transformation processes the relational input from the upstream transformation and provides JSON output to the downstream transformation. The Hierarchy Builder transformation produces hierarchical output based on the JSON schema definition file that you associate with the transformation.

1. Drag the Hierarchical Builder transformation from the **Transformation** palette into the editor.
2. Click the **Properties** panel to configure the properties.

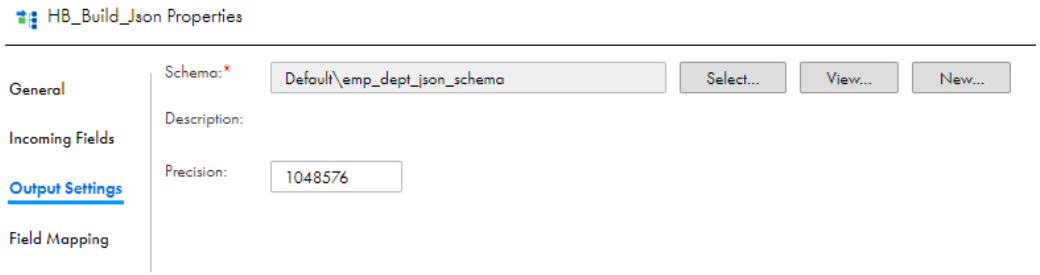
3. Establish a connection between the Source and the Hierarchical Builder transformation.

Associate the Hierarchical Schema with the Hierarchical Builder Transformation

Select the sample schema on the **Output Settings** tab of the **Properties** panel.

1. Select the Hierarchical Builder transformation and click the **Output Settings** tab under the **Properties** panel.
2. Select the schema you created to convert the relational input into the hierarchical output.

   The following image shows the **Output Settings** tab:



3. To save the schema and include it in the Hierarchical Builder transformation, click **Save**.

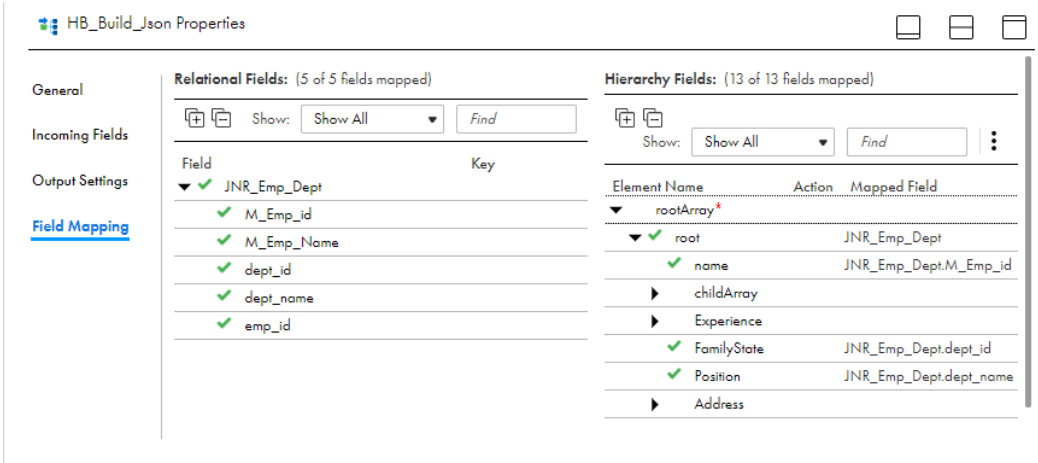
Configure the Field Mapping to Select Schema Elements

Configure the field mapping in the Hierarchy Builder transformation to define the relationship between the relational elements and schema elements. You use the field mapping editor to link relational fields to schema elements to provide the hierarchical output.

- In the **Properties** panel of the Hierarchical Builder transformation, click the **Field Mapping** tab.

  The field mapping editor displays the transformation input schema elements on the left side and the relational output fields on the right side of your screen.

  The following image shows the field mapping editor:



Add a Target Transformation

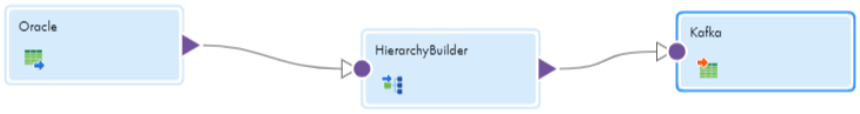
Use the Target transformation to define the Kafka target in the mapping.

1. Drag a Target transformation onto the mapping canvas.
2. Click the **Target** tab under the **Properties** panel.
3. Select an active Kakfa connection and select a Kafka target object.
4. Select `None` as **Format Type** in **Formatting Options**.
5. Establish a connection between the Hierarchy Builder transformation and the target.
6. Click **Save**.

Configure a Mapping Task

You can save and run the mapping directly. You can also choose to configure a mapping task, add the configured mapping, and run the mapping task.

1. To create a mapping task, click **New** > **Tasks**, and then click **Create**.
2. In the **New Mapping Task** window, provide a task name, and provide the configuration details.
3. Select the configured mapping, and click **OK**.
1. The following image shows the configured mapping:



4. Configure a schedule and advanced options, if required.
5. Click **Save**.

Author

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Acknowledgements

The author would like to acknowledge Sarvesh Jaiswal for his technical assistance with this article.